



SYS_READY // D1-TRACK

AeroReady

Mission Readiness & Predictive Maintenance Copilot

From sensor evidence to mission-readiness decisions.

TEAM ASTRAEA

IBM Bob Hackathon — Defense & Aerospace Track D1: Mission Readiness & Predictive Maintenance

Readiness Data Exists. A Readiness Answer Doesn't.

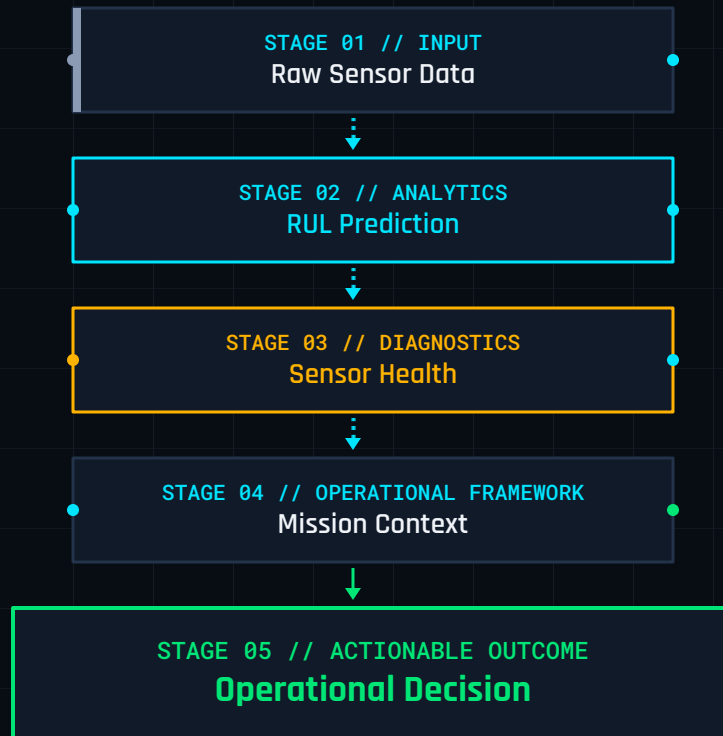
Military and aerospace organizations possess large volumes of sensor and service information, but decision-makers lack a single, timely assessment of mission readiness.

Challenge: Determine readiness now, rather than simply predicting component failure.

KEY OPERATIONAL QUESTIONS

Is an asset ready for the planned mission window?
How much Remaining Useful Life (RUL) remains?
Does sufficient safety margin exist?
Which sensors show abnormality or degradation?
Which assets require maintenance first?
Why is a particular asset not mission-ready?

READINESS ASSESSMENT FLOW



// CORE DECISION PIPELINE

01 Data Ingestion & Validation

Continuous ingestion of raw multi-sensor telemetry and structured CSV data streams.

02 LSTM RUL & Health Analysis

LSTM RUL prediction combined with Isolation Forest anomaly detection for asset health.

03 Mission Readiness Engine

Calculates operational Safety Margin, Risk factors, and real-time Readiness Score.

04 Maintenance Priority & Copilot

Outputs clear priority actions via an evidence-grounded operational assistant.

// OPERATIONAL GOVERNANCE PRINCIPLES

1. Measurable Technical Evidence

The machine-learning layer produces precise, quantitative technical metrics rather than non-deterministic estimates.

2. Operational Risk Assessment

The mission-readiness layer directly converts raw algorithmic evidence into actionable operational assessments.

3. Zero-Hallucination Explanations

Finally, the Copilot explains each assessment using strictly stored backend telemetry to prevent unsupported AI claims.

SYSTEM ARCHITECTURE DIAGRAM

From CSV to Mission Decision

STAGE 01

Data Layer

[INPUT FLOW]

- CSV Dataset
- Telemetry Stream

[PROCESSING]

- Upload API
- Data Parser

[STORAGE]

- Dataset Repository

STAGE 02

Predictive Pipeline

[ROUTING]

- Pipeline Service

[ML MODELS]

- **LSTM RUL Predictor**
- Isolation Forest Health Analysis

[OUTPUT]

- Persisted Results Store

CORE ENGINE

Mission Decision

[EVALUATES]

- Predicted RUL
- Mission Duration
- Safety Margin
- Sensor Health

[DECISION OUTPUT]

- **Risk Flag**
- **Readiness Score**
- Recommendation

STAGE 04

Application

[API FRAMEWORK]

- Flask API Backend

[REACT DASHBOARD]

- Fleet View
- Sensor Health
- Maintenance
- **AI Copilot View**

STAGE 05

Integration

[AGENTIC TOOLS]

- **IBM Bob Engine**
- MCP Tools Ecosystem

[CAPABILITY]

- Live Tool-Based Investigation
- Automated Telemetry Querying

Two Models. One Operational Decision.

LSTM RUL PREDICTOR

- **Sequence Length:** 30 cycles of 14 smoothed sensor features.
- **Architecture:** Hidden size 64, 2 LSTM layers, 0.2 dropout, outputting a scalar RUL.

ISOLATION FOREST HEALTH ANALYSIS

- **Configuration:** 100 estimators, 0.01 contamination.
- **Rules:** 5-cycle persistence window, 3.0σ deviation threshold. Identifies NORMAL, DEGRADING, ABNORMAL, or UNKNOWN states.

MISSION READINESS ENGINE

SAFETY MARGIN EQUATION

$$\text{MARGIN} = \text{PREDICTED RUL} - \text{MISSION DURATION}$$

PREDICTED RUL

42.3

MISSION DUR.

30.0

=

MARGIN

+12.3

READINESS ASSESSMENT

MARGINAL



0 CYCLES MARGINAL THRESHOLD 50+ CYCLES

- **Telemetry Warning:** 2 degrading sensors identified

SYSTEM RECOMMENDATION

Recommend physical maintenance inspection before aircraft dispatch.



OPERATIONAL DASHBOARD USER JOURNEY

ONE ASSET. ONE MISSION. ONE EVIDENCE-BASED DECISION.

1

PHASE 01

FLEET OVERVIEW

View overall fleet readiness metrics and high-level maintenance priorities across assets.

STATUS: OPTIMAL

2

PHASE 02

ASSET SELECTION

Select a specific high-priority turbine engine requiring immediate inspection.

TARGET: ENG-2491

3

PHASE 03

MARGIN ASSESSMENT

Inspect the Predicted RUL (Remaining Useful Life) against Mission Safety Margin.

DELTA RUL: -14 CYC

4

PHASE 04

SENSOR HEALTH

Open the Sensor Health tab to inspect real-time degradation evidence & telemetry.

SENSORS: 4 DEGRADED

5

PHASE 05

RECOMMENDATION

Review automated actionable operational and maintenance recommendations.

ACTION: HOLD FLIGHT

6

STEP 06 // INTERACTIVE COPILOT

QUERY COPILOT INTERFACE

OPERATOR PROMPT:

"Why is this engine not mission ready?"

ASSET: ENG-2491

MODEL: RUL-LSTM-V2

NL-QUERY ACTIVE

Natural language interface allows operators to query system diagnostics directly.

7

STEP 07 // EVIDENCE CALLOUT & DECISION

GROUNDED RESPONSE & BACKEND EVIDENCE CALLOUT

SYSTEM RESPONSE (VERIFIED BACKEND EVIDENCE):

"Engine ENG-2491 predicted RUL is **42 cycles**, falling short of the required mission threshold of **56 cycles**."

Primary driver: Accelerated HPC outlet temperature sensor degradation (Sensor T50 drift +1.8%) and acoustic vibration anomalies in Turbine Stage 2.

100% GROUNDED IN STORED TELEMETRY & HISTORICAL MAINTENANCE LOGS

IBM Technology Integration: IBM Bob + MCP

The IBM Bob integration uses the Model Context Protocol (MCP) to expose AeroReady's read-only data as queryable tools.

WORKFLOW ARCHITECTURE END-TO-END EXECUTION FLOW



DOCUMENTED MCP TOOLS

- `get_fleet_summary`
- `get_engine_readiness`
- `get_sensor_health`
- `list_high_priority_engines`
- `ask_copilot`

IMPLEMENTATION STATUS METRICS

FULLY IMPLEMENTED CAPABILITIES

AeroReady core analytics, ML pipeline, Flask API, and MCP endpoint definitions are fully deployed and validated.

REMAINING IMPLEMENTATION TARGET

The full IBM Bob MCP integration deployment represents the designed remaining implementation phase.

Technical Evidence & Explainability

Audited Model Evaluation Metrics

MAE

10.05

RMSE

13.38

R² SCORE

0.897

Engine 1 Analysis Case Study

LIVE TELEMETRY & EVALUATION READOUT

Telemetry Data Inputs

RUL: 116.44 cycles | Sensor Health: 14/14 NORMAL

Mission Plan: 30 cycles

Automated Assessment

Status: SAFE | Safety Margin: 86.44 cycles

READINESS SCORE: 93.15 / 100

Trust & Governance Safeguards

DETERMINISTIC VERIFICATION FRAMEWORK

Every factual statement is derived strictly from stored backend evidence.

The architecture maintains zero-trust execution bounds, ensuring operational telemetry aligns precisely with military-grade safety audit requirements.

Zero Hallucination Guarantee

The system strictly prohibits invented failure probabilities or fabricated maintenance history.

No Component Fabrication

Prevents hallucinated component failures by binding reasoning directly to sensor ground truth.

Immutable Engine Overrides

The Copilot cannot override the readiness engine and explicitly reports when evidence is insufficient.

From One Engine to Fleet-Level Mission Readiness

Capabilities Today

CURRENT CORE ENGINE ANALYTICS

- Predict Remaining Useful Life
- Detect multivariate sensor degradation
- Evaluate mission readiness
- Prioritize maintenance
- Explain the evidence

Vision Beyond the Hackathon

SCALING & PRODUCTION ROADMAP

- Integration of aircraft-specific validated data and maintenance records.
- Implementation of calibration procedures and domain safety approval.
- Deployment of fleet-scale predictive maintenance and production database infrastructure.

[SAFETY BOUNDARY DIRECTIVE]

AeroReady predictions are designed to serve strictly as decision support for maintenance and operations teams, not as autonomous flight-safety commands.



AEROREADY

PROJECT CLOSING & TEAM OVERVIEW

Team Astraea & Closing

Krish Parmar

EDA / DATA ANALYSIS + BACKEND

Mayur Maghrola

FRONTEND + INTEGRATION

Deep Kayastha

BACKEND + AI / ML

Dhairya Patel

BACKEND + DATABASE

CORE MISSION STATEMENT

"AeroReady does not just predict failure; it determines whether an asset has enough remaining life for the mission, identifies the evidence behind the risk, and helps the operator decide what needs attention first."

AeroReady turns raw aerospace sensor evidence into a traceable mission-readiness decision.